

1RV24MC042_HARSHAVARDHAN_BR_LAB9

Organize Kubernetes Pods Using Labels and Namespaces

File structure

```
Program-9/  
├── namespaces.yaml  
└── pods.yaml
```

```
> minikube start  
🐳 minikube v1.37.0 on Linuxmint 22.1  
🔧 Using the docker driver based on existing profile  
👉 Starting "minikube" primary control-plane node in "minikube" cluster  
📦 Pulling base image v0.0.48 ...  
🔄 Updating the running docker "minikube" container ...  
  
🚨 Docker is nearly out of disk space, which may cause deployments to fail (due to capacity). You can pass '--force' to skip this check.  
💡 Suggestion:  
  
Try one or more of the following to free up space on the device:  
  
1. Run "docker system prune" to remove unused Docker data (optionally with "--volumes").  
2. Increase the storage allocated to Docker for Desktop by clicking on: Docker icon > Preferences > Resources > Disk Image Size  
3. Run "minikube ssh -- docker system prune" if using the Docker container driver.  
  
🍿 Related issue: https://github.com/kubernetes/minikube/issues/9024  
  
🐳 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...  
🔍 Verifying Kubernetes components...  
▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0  
▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5  
▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8  
▪ Using image docker.io/kubernetesui/dashboard:v2.7.0  
^[[3~💡 Some dashboard features require the metrics-server addon. To enable these features please run:
```

Namespaces.yaml

```
apiVersion: v1  
kind: Namespace  
metadata:  
  name: dev  
---  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: prod
```

pods.yaml

```
apiVersion: v1  
kind: Pod  
metadata:  
  name: frontend-pod  
  namespace: dev  
  labels:  
    app: frontend  
    env: dev  
spec:
```

```

containers:
  - name: frontend
    image: nginx
    ports:
      - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80

```

Step 3 : Apply configuration

```

kubectl apply -f namespaces.yaml
kubectl apply -f pods.yaml

```

Step 4 : Get pods wrt Namespace

```

kubectl get pods -n dev
kubectl get pods -n prod

```

```
> kubectl apply -f namespaces.yaml
kubectl apply -f pods.yaml
```

```
namespace/dev created
namespace/prod created
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
> kubectl get pods -n dev
kubectl get pods -n prod
NAME          READY   STATUS             RESTARTS   AGE
backend-pod   1/1     Running             0           7s
frontend-pod  0/1     ContainerCreating   0           7s
NAME          READY   STATUS             RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0           7s
frontend-pod  0/1     ContainerCreating   0           7s
```

Step 5: Check All Pods

```
kubectl get pods --all-namespaces
```

Step 6: Filter Pods Using Labels

```
> kubectl get pods --all-namespaces
NAMESPACE     NAME                                READY   STATUS    RESTARTS   AGE
demo          nginx                              1/1     Running   1 (3d10h ago)  3d13h
dev           backend-pod                        1/1     Running   0           51s
dev           frontend-pod                      1/1     Running   0           51s
kube-system   coredns-66bc5c9577-sxnnl         1/1     Running   7 (3d10h ago)  17d
kube-system   etcd-minikube                     1/1     Running   7 (3d10h ago)  17d
kube-system   kube-apiserver-minikube           1/1     Running   7 (3d10h ago)  17d
kube-system   kube-controller-manager-minikube  1/1     Running   7 (3d10h ago)  17d
kube-system   kube-proxy-9jhf7                  1/1     Running   7 (3d10h ago)  17d
kube-system   kube-scheduler-minikube           1/1     Running   7 (3d10h ago)  17d
kube-system   metrics-server-85b7d694d7-nj8kc  1/1     Running   10 (3d10h ago)  17d
kube-system   storage-provisioner               1/1     Running   15 (71s ago)   17d
kubernetes-dashboard dashboard-metrics-scraper-77bf4d6c4c-259hn 1/1     Running   7 (3d10h ago)  17d
kubernetes-dashboard kubernetes-dashboard-855c9754f9-8sw95 1/1     Running   14 (71s ago)   17d
prod          backend-pod                        1/1     Running   0           51s
prod          frontend-pod                      1/1     Running   0           51s
```

Step 7: List all frontend pods in dev

```
kubectl get pods -n dev -l app=frontend
```

```
> kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           74s
```

```
> kubectl get namespace
NAME                STATUS   AGE
default             Active   17d
demo                Active   3d13h
dev                 Active   108s
kube-node-lease     Active   17d
kube-public         Active   17d
kube-system         Active   17d
kubernetes-dashboard Active   17d
prod               Active   108s
```

Step 8 : Cleanup

```
kubectl delete pods --all -n dev
kubectl delete pods --all -n prod
```

```
kubectl delete namespace dev prod
```

```
> kubectl delete pods --all -n dev
kubectl delete pods --all -n prod

kubectl delete namespace dev prod
pod "backend-pod" deleted from dev namespace
pod "frontend-pod" deleted from dev namespace
pod "backend-pod" deleted from prod namespace
pod "frontend-pod" deleted from prod namespace
namespace "dev" deleted
namespace "prod" deleted
```